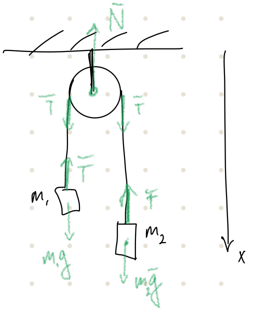
Ньютон



$$\begin{cases} m_2a = m_2g - T & \mid : m_2 \\ -m_1a = m_1g - T & \mid : m_1 \end{cases} \oplus$$

$$a = \frac{(m_2 - m_1)g}{m_1 + m_2}$$

$$a = g - T/m_1$$
$$-a = g - T/m_2$$

$$2g = T\left(\frac{1}{m_1} + \frac{1}{m_2}\right)$$

$$T = \frac{2m_1m_2}{m_1 + m_2}g$$

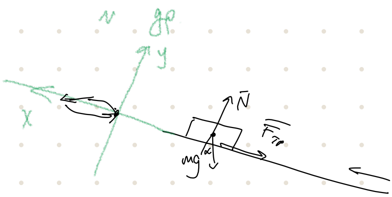
$\bar N = 2\bar T$ по 3 з. Ньютона

$$N = 2T = \frac{4m_1m_2}{m_1+m_2}g$$

$$(m_1+m_2)g \geqslant \frac{4m_1m_2}{m_1+m_2}g$$

$$m_1^2 + 2m_1m_2 + m_2^2 \geqslant 4m_1m_2$$

$$(m_1 - m_2)^2 \geqslant 0$$



$$t_{подъема} = ? \, t_{спуска}$$

$$a_c = g(m\alpha - \mu\cos\alpha)$$

$$a_п = g(m\alpha + \mu\cos\alpha)$$

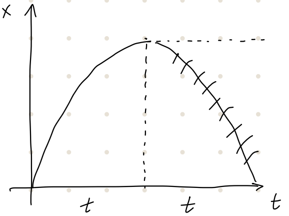
$$N - mg\cos\alpha = 0$$

$$ma = -F_{тр} - mg\sin\alpha$$

$$F_{тр} = \mu N$$

$$ma = -\mu mg\cos\alpha - mg\, m\alpha$$

$$S = S_0 + V_0t + \frac{at^2}{2}$$



$S = a_п t^2/2$

$S = \frac{a_c t^2}{2}$

$a_п t_п^2 = a_c t_c^2$

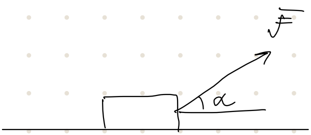
Т.о. удобно считать $t_{подъема}$ дополн. параболу до $t_{спуска}$

$\eta^2 a_п = a_c$

$\eta g(\sin\alpha + \mu\cos\alpha) = g(\sin\alpha - \mu\cos\alpha)$

$\mu = \frac{(\eta^2 - 1)}{\eta^2 + 1} \operatorname{tg}\alpha$

№ др.



μ

$\alpha(F_{min}) - ?$

F_{min} ?

$\alpha = 0: \quad F = \mu mg$

$\alpha = \frac{\pi}{2}. \quad F = mg$

$\alpha = \operatorname{arctg}\mu$

$\sin^2\alpha + \cos^2\alpha = 1$

~~$\cos^2\alpha = 1 - \sin^2\alpha$~~ $\quad \operatorname{tg}^2\alpha = -1 + \frac{1}{\cos^2\alpha}$

$\cos^2\alpha = \frac{1}{1 + \operatorname{tg}^2\alpha} = \frac{1}{1 + \mu^2}$

$\sin^2\alpha = \frac{\mu^2}{1 + \mu^2}$

$F = \frac{\mu mg}{\sqrt{1/(1+\mu^2)} + \mu\sqrt{\mu^2/(1+\mu^2)}} = \frac{\mu mg}{\sqrt{\frac{1}{1+\mu^2}}(1+\mu^2)} = \frac{\mu mg}{\sqrt{1+\mu^2}} \geq 1$

$\frac{\mu}{\sqrt{1+\mu^2}} \leq \mu mg$

$$\frac{\mu}{\sqrt{1+\mu^2}} < 1$$

$$\mu < \sqrt{1+\mu^2}$$

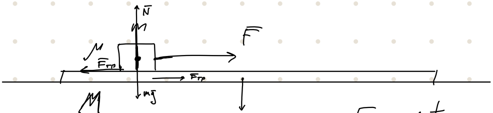
$$\mu^2 < 1+\mu^2$$

$$F = \frac{\mu mg}{\cos\alpha - \mu\sin\alpha}$$

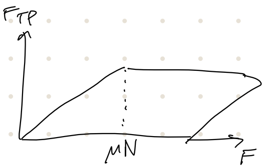
$$\cos\alpha = \mu\sin\alpha$$

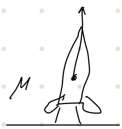
$\mu \leq ctg\,\alpha$ ← если при таком вниз толкая вообще не сдвинешь.

<u>№ др.</u>



$F = \alpha t$





$F = \alpha t$

T

на какой высоте взорв. рак.

$a_т(t)$

$(m+M)a = F = \alpha t$ ← сила тр. покоя — тела движ. вместе

$$a = \frac{\alpha}{M+m}t$$

$$\begin{cases} m a_1 = F - F_{тр} \\ M a_2 = F_{тр} \end{cases} \rightarrow \begin{cases} m a_1 = F - \mu mg \\ M a_2 = \mu mg \end{cases} \quad \begin{cases} a_1 = \frac{F}{m} - \mu g = \frac{\alpha t}{m} - \mu g \\ a_2 = \frac{\mu mg}{M} \end{cases}$$

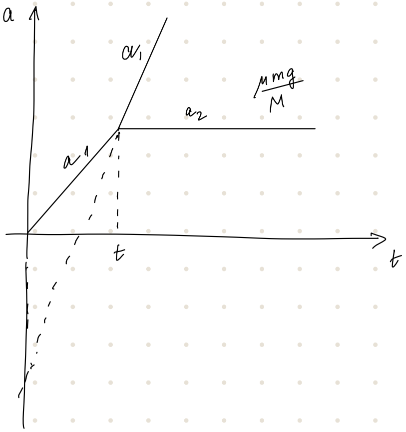
$$F_{тр} = \frac{M}{m+M}\alpha t$$

сила тр. покоя $\cancel{=}$ всегда < силы N

Когда начин. скольз. $a_1 = a_2$: $\frac{\alpha t}{m} - \mu g = \frac{\mu mg}{M}$

одно и то же

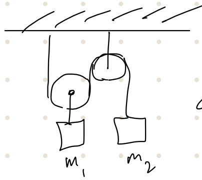
$$t = \frac{m}{\alpha}\left(1 + \frac{m}{M}\right)\mu g$$



Д/З: 1) греческий алфавит

2) про рокету

3)



a - ?

4) 1,69

5) 1,81

6) 1,84